

Vansh Gupta

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EDUCATION

- Indian Institute of Technology, BHU, Varanasi
Bachelor + Master of Technology - Engineering Physics;

Dec 2021 - June 2026
CGPA-8.14/10

AREAS OF INTEREST

Natural Language Processing, Graph Machine Learning, Representation Learning, Reinforcement Learning, Retrieval Augmented Generation, LLM finetuning and Interpretability/Explainability.

PUBLICATIONS/PRE-PRINTS

- Enriching Pre-Training using Fuzzy Logic.** Vansh Gupta, Vandana Bharti, Abhinav Kumar, Anshul Sharma and Sanjay Kumar Singh. *IEEE international conference of Fuzzy Systems (Fuzz-IEEE) 2025.*
- Leveraging Diagnostic Scores To Improve Depression Classification From Screening Interviews.** Tanmay Surana, Vansh Gupta, Chng Eng Siong.
- Time in Residual Stream: How Language Models Encode Temporal Horizon.** Vansh Gupta, Justin Shenk. (Under submission in *NeurReps, Interpreting Agent Behaviour and Interp4Discovery Workshops*, NeurIPS 2026.

EXPERIENCE

Research Fellow - Supervised Program for Alignment Research (SPAR). March 2026 - May 2026 *Causal Intervention, Representation Geometry Analysis, Latent Space Dynamics, Activation and Attribution Patching*

- Generated 14.7K-triple synthetic corpus and 543 graded scenarios spanning 1 day-500 years using Qwen3-32B on vLLM with closed-vocabulary anchors, banned-word constraints, and ± 2 -token length matching.
- Trained layer-wise linear probes on residual streams of 7 LLMs (1.7B-14B, three families) via TransformerLens, reaching 99% accuracy; performance validated with shuffle, random-direction, and low-rank controls.
- Established log-linear (Weber-Fechner) horizon encoding by fitting five functional forms (power-law, linear-in-rank, piecewise, sigmoid) via AIC/BIC with bootstrap CIs; log-linear won 97.4% of fits, $R^2=0.934$.
- Quantified manifold geometry via PCA with cubic-spline tortuosity, persistent homology (Vietoris-Rips, Wasserstein/bottleneck, Betti), and TwoNN intrinsic dimension; corrected a $192\times$ activation-norm confound.
- Ran full and direction-restricted activation patching with per-head circuit attribution, exposing an $\text{encode}\neq\text{use}$ dissociation; benchmarked linear against manifold-spline steering using Bhattacharyya energy.

Research Intern - IBM Research, Bangalore, India.

May 2025 - Aug 2025

Exposure: Static Code Analysis, Synthetic Data Creation, Repository Level Code Analysis, Chain-Of-Thought prompting.

- Informed model development by conducting a comprehensive literature review of **code chunking**, **Retrieval Augmented Code Generation**, **multimodal embedding models** and **contrastive learning** techniques.
- Manually engineered a dataset for repository-level code comprehension (**25** multi-file, **25** single-file questions) and benchmarked SOTA embedding models(+**integrating callgraph** of the repository) to identify top performers.
- Designed a threefold **Agentic** pipeline with **LLM-as-a-judge** evaluation setup for creating a novel dataset of query-file(s) tuples addressing the **repository level code capability comprehension** and understanding.

Research Intern - Language and Speech Lab, NTU Singapore

Oct 2024 - July 2025, Feb 2026 - Jul 2026

Exposure: LLM fine-tuning, PEFT, LoRA, LLaMA Adapters, Emotion Extraction from Language, Prompt Tuning

- Researched and implemented advanced **LLM fine-tuning** strategies (PEFT, LoRA, Adapters, Prompt Tuning) on Lightning.AI's litgpt to enhance **emotion extraction** and depression detection in low-resource settings.
- Experimented with **data augmentation**, **attention-based models**, and **adversarial perturbation techniques**, integrating robust loss functions such as **focal loss** to improve generalization and model resilience.
- Achieved **69% F1 score** on the DAIC-WOZ dataset, beating State-of-the-Art embedding models.
- Fine-tuned **Qwen3-8B** and **Qwen3.5-9B** on COKE dataset using **LORA** and evaluated the performance using **BLEU**, **METEOR**, **ROUGE** metrics for open ended text generation and Emotion Classification.

Research Intern - Visual Computing and Data Analytics Lab, IIT BHU

Aug 2023 - April 2024

Exposure: GraphMamba, GANs, Graph Neural Networks, Fuzzy Logic Systems, DeepWalk, Node2Vec, Struc2Vec

- Redesigned the **GraphGAN** model with **Wasserstein Loss**, increasing the accuracy from **84.7% to 88.57%**.
- Implemented DeepWalk, Node2vec, Struc2Vec to generate pre-trained node embedding vectors. Link
- Pioneered a **Fuzzy Pre-Processing layer** with **modified K-Means Clustering** to extract semantic features from node embeddings, boosting downstream **link-prediction** accuracy with **GraphMamba** by up to 5%. Link

PROJECTS

IOI circuit detection - Interpretability in the wild

Exposure: Mechanistic Interpretability, activation patching, circuit discovery, logit lens analysis, causal ablations

- Reimplemented the Interpretability in the Wild study by reproducing end-to-end mechanistic analysis of **Indirect Object Identification (IOI)** in GPT-2 Small, including circuit identification at the attention-head level.
- Implemented IOI data generation, activation patching, and **template-conditioned mean ablations** to causally isolate attention heads and trace information flow through the residual stream using **path patching**.
- Identified, categorized, and validated functional head classes (e.g., Name Movers, S-Inhibition, Induction, Duplicate Token heads) and evaluated **faithfulness**, **completeness**, and **minimality** via knockouts. [Link](#)

Understanding ModernLLM and Alignment

Exposure: Mixture-of-Experts, sliding window attention, rolling buffer KV Cache, RLHF, Curriculum Learning

- Engineered core LLM components from scratch, including **RMSNorm**, **Rotary Positional Embeddings (RoPE)**, **SwiGLU** activations, **rolling buffer KV caching**, **sliding window Attention** and **Mixture-of-Experts (MoE)** layers for scalable and efficient transformer architectures.
- Developed pipelines for **Supervised Fine-Tuning (SFT)**, **instruction-based data formatting**, and **reward modeling** using pairwise preference datasets with **Bradley-Terry** and **margin ranking** objectives.
- Implemented Reinforcement Learning from Human Feedback (**RLHF**) techniques using **PPO** and **GRPO**, incorporating **reward normalization**, **KL regularization**, and **group-relative advantage estimation** for stable policy optimization. [Link](#)

PaliGemma-Reimplementation

Exposure: Multi/Grouped Query Attention, KV Cache, MultiModal LMs, Vision Transformers, CLIP, RoPE

- Reimplemented **google/siglip-so400m-patch14-384** and **google/gemma-2b** architectures from scratch, gaining hands-on experience with large-scale **Vision Transformers** and **Language Models**.
- Engineered a **Multimodal Language Model** by designing a linear projector to align image and text embeddings, enabling effective cross-modal representation learning.
- Performed inference with **KV Cache** and applied **top-p sampling**, both implemented from scratch, using pre-trained HuggingFace weights, improving efficiency and controllability of generation. [Link](#)

ACHIEVEMENTS AND EXTRACURRICULAR

- : **AIR 6332 in JEE Mains 2021 (Top 99.40% percentile)**, **AIR 7196** in IIT-JEE Advanced 2021
- : **Top 1%** and **State Topper** in Indian Olympiad Qualifier for Chemistry 2020 among 17,000+ participants
- : Secured a **scholarship** for the Bertelsmann Next Generation Tech Booster – Udacity Nanodegree in Generative AI

RELEVANT COURSES

- **Academic Courses::** Fuzzy Set Theory, Linear Algebra, Differential Equations, Probability and Statistics, Complex Analysis, Wavelet Analysis, Operations Research
- **Online Courses(MOOC)::** Deep Generative Modelling(Micheal Elad, Technion), Large Language Models(LCS2, IITD), Introduction to Reinforcement Learning(NPTEL IITM), Interpretability of LLMs (Tel Aviv University)

COMMUNITY ENGAGEMENT

- **Students' Parliament::** Member of Undergraduate Academic Affairs Committee. Worked with a team of 52 other parliament members and put forward 5 proposals in one academic year.
- **Training and Placement Cell::** Worked with a team of 30+ other representatives. Managed interview processes of 10+ companies and open assessments of 40+ companies during the internships and placements.
- **Astronomy Club-Senior Core Team Member::** conducted observational sessions for 100+ students, a workshop on "Introduction to Machine Learning" in collaboration with GDSC and organized a Data Analytics competition, "Stellar Analytics", with 10+ participating teams.

SKILLS SUMMARY

- **Languages:** Python, C, C++, SQL, Bash
- **Deep Learning:** PyTorch, Tensorflow, PyTorch Lightning, Hugging Face, PyTorchGeometric, TensorBoard
- **Data Science:** Scikit-Learn, Langchain, LangGraph, Gensim, StatsModels, Pandas, Numpy, Matplotlib, PySpark, FAISS
- **MLOPS Tools:** GIT, GitHub Actions, DVC, MLFlow
- **Platforms:** Linux, Web, Windows

RECOMMENDATIONS

- **Prof. Chng Eng Siong, Nanyang Technological University, Singapore:** [Link](#)